set_pg_via_master_rule

NAME

set_pg_via_master_rule
Creates a new via master rule for the power and ground network. The via rule is defined by the contact code, location of contact code inside the intersection box, cut spacing, multiplication, and so on.

SYNTAX

```
status set_pg_via_master_rule
        rule_name
        [-contact_code via_contact_code]
        [-via_array_dimension {column row}]
        [-offset_start center | lower_left ]
        [-offset {x_offset y_offset}]
        [-orient R0 | R90]
        [-allow_multiple {x_pitch y_pitch}]
        [-cut_spacing {x_spacing y_spacing}]
        [-via_site_ratio {x_ratio y_ratio}]
        [-track_alignment alignment_mode]
        [-cut_mask mask_one | mask_two | mask_three]
        [-snap_reference_point {x y}]
```

Data Types

```
rule_name          string
via_contact_code   string
column             integer
row                integer
x_offset           float
y_offset           float
x_pitch            float
y_pitch            float
x_spacing          float
y_spacing          float
x_ratio            float
y_ratio            float
alignment_mode     string
x                  float
y                  float
```

ARGUMENTS

rule_name
Specifies the name of the power ground via rule. This option is required.

-offset {x_offset y_offset}
Specifies the horizontal (x_offset) and vertical (y_offset) values to use when creating a contact code inside the intersection box. The units are um. The offset is applied with respect to the center of the intersection box. By default, the offset is 0.

-offset_start center | lower_left
Specifies the offset reference point in the intersection box. Keyword center refers to the center of the intersection box. Keyword lower_left refers to the lower-left corners of the intersection box.

-contact_code via_contact_code
Specifies the via master or contact code used to create the via rule. By default, the tool uses the default contact codes.

-via_array_dimension {column row}
Specifies the via array dimension size by integer values for column and row. This option can only be applied to a simple symmetric contact code. The tool creates the via array based on the specified contact code. If the contact code is not symmetric, this option is ignored. By default, column and row are calculated based on the size of the intersection, and the tool fills the intersection with via cuts.

-orient R0 | R90
Specifies the orientation of the contact code by keywords R0 and R90 respectively. By default, the via has orientation R0 and is not rotated.

-cut_mask mask_one | mask_two | mask_three

Specifies the cut mask for this via master rule by keywords mask_one, mask_two or mask_three. By default, the via cut mask is not colored. Via metal enclosure masks follow PG wire masks.

-allow_multiple {x_pitch y_pitch}
Specifies the horizontal (x_pitch) and vertical (y_pitch) values when filling an intersection with multiple repetitions of the contact code or via array. The unit is defined in the technology file. If the via array is created with the -via_array_dimension option, the via array is repeated. Otherwise, the contact code defined by the -contact_code option is repeated. By default, the contact code or via array is not repeated.

-cut_spacing {x_spacing y_spacing}
Specifies the horizontal (x_spacing) and vertical (y_spacing) values between cuts in a simple array via. The spacing values are ignored if smaller than the minimum cut spacing. The unit of cut spacing is in um. By default, the default cut spacing is used.

-via_site_ratio {x_ratio y_ratio}
Specifies the via site horizontal and vertical ratio by x_ratio and y_ratio. The valid range for ratio is from 0.0 to 1.0 while 0.0 is invalid. By default, the ratio is 1.0 and the whole intersection is used for via creation. When specified, the intersection box is shrunk by the specified ratio for via creation.

-track_alignment alignment_mode
Specifies the track alignment option for this via master rule. Valid values are track, half_track, top_track_bottom_half_track, top_half_track_bottom_track, top_track_only, top_half_track_only, bottom_track_only, and bottom_half_track_only.

By default vias created by this rule do not align to routing track or half routing track. The top and bottom of a via can be aligned to track or half-track respectively. The alignment can be applied to both of layers or just one of the top and bottom layers.

-snap_reference_point {x y}
Specifies the reference point for repeating vias. This option needs to be used together with the "-allow_multiple" option for repeating vias. When specified, the origin of the first via or the lower-left via created inside the intersections with this via master rule will be snapped to the specified point, or the specified point plus a multiple of pitch values if the reference point is outside the intersection. The "-offset_start" or "-offset" options cannot be used together with this option. If not specified, repeating vias will honor offset options or get evenly distributed inside intersection bounding boxes.

DESCRIPTION
Specifies the power ground via rules such as the location of contact code inside the intersection box, orientation and multiplication etc. The PG via rule can be used during pattern definition and when creating the power and ground network.

EXAMPLES
The following example defines a power ground via rule with the name via_rule_1 using contact code VIA34. The offsets are 3 um and 4 um with respect to the center of the intersection. The via array size is 5x5.

```
  prompt> set_pg_via_master_rule \
     via_rule_1 -contact_code VIA34 \
     -offset {3 4} -via_array_dimension {5 5} \
```

MORE EXAMPLES
For more example, please start GUI and invoke the following command.
gui_show_task -task "Design Planning:PG Planning->Examples->Overview"

SEE ALSO
compile_pg(2)
create_pg_composite_pattern(2)
create_pg_macro_conn_pattern(2)
create_pg_ring_pattern(2)
remove_pg_via_master_rules(2)
report_pg_via_master_rules(2)